

Remittance Outbound API Specifications

Version 1.0.0

Herein is the specification of a set of APIs that provides functionality to send money out of the country via selected providers.

APIs:

- Quotation /Book Payment API
- Transaction Confirm API
- Status Check API
- Country Names / Codes lookup API
- Bank Names / Codes lookup API
- Wallet Names / Codes lookup API

1.0 Quotation / Book Payment API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/quote

Request Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
reference_number	String	Conditional	Transaction reference number	50
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
source.currency	String	Mandatory	Source of funds currency	3
source.amount	String	Mandatory	Amount	10
source.accountNumber	String	Mandatory	Debit account	20
destination.countryCode	String	Mandatory	Country Name or Country Code	3
destination.bicWallet	String	Mandatory	Destination Bank BIC or Wallet	100
destination.currency	String	Mandatory	Source of funds currency	3

destination.amount	String	Optional	Converted to receiving currency and net amount of charges	10
destination.accountNumber	String	Mandatory	Credit account	50
channel	String	Mandatory	HTTP header	100
signature	String	Mandatory	Signature based from(rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber destination.BIC.wallet and channel	

Response: JSON
Response Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
reference_number	String	Conditional	Transaction reference number	50
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment	10
source.currency	String	Mandatory	Source of funds currency	3
source.amount	String	Mandatory	Amount	10
source.accountNumber	String	Mandatory	Debit account	20
destination.currency	String	Mandatory	Source of funds currency	3
destination.amount	String	optional	Converted by the provider	10
destination.accountNumber	String	Mandatory	Credit account(Account or Mobile)	50
status	String	Mandatory	POSTED PENDING REJECTED	20

responseCode	String	Mandatory	Any response or error code provided by Provider	20
responseDesc	String	Optional	If failure reason or addition date	300
transactionId	String	Conditional	If posted successful the Provide transaction id	50
destination.BICWallet	String	Mandatory	Destination Bank or Wallet	300
channel	String		HTTP header	100
charge.amount	String	Mandatory	Amount deducted from the Principal Amount	10
charge.currency	String	Mandatory	The currency of charged Amount	3
signature	String	Mandatory	Signature based from(rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber, destination.BIC.wallet and channel	

2.0 Transaction Confirm API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/transaction

Request Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
reference_number	String	Conditional	Transaction reference number	50
quotationId	String	Mandatory	Transaction Id returned in Quotation response	
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50

transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
source.currency	String	Mandatory	Source of funds currency	3
source.amount	String	Mandatory	Amount	10
source.accountNumber	String	Mandatory	Debit account	20
sender.firstName	String	Mandatory	Sender's Name max 50	100
sender.middleName	String	Optional	Senders Middle Name	100
sender.lastName	String	Mandatory		100
sender.mobileNumber	String	Optional	Mobile Number of the sender	12
sender.email	String	Optional	Email of the sender	200
sender.address.countryCode	String	Optional		3
sender.address.postalcode	String	Optional		50
sender.address.line1	String	Optional		300
sender.address.line2	String	Optional		300
sender.address.city	String	Optional		200
destination.currency	String	Mandatory	Source of funds currency	3
destination.amount	String	optional	Converted by the provider	10
destination.accountNumber	String	Conditional Mandatory for payments to Accounts	Credit account	50
destination.bicWallet	String	Mandatory	Destination Bank or Wallet	300
receiver.firstname	String	Mandatory	First name of the Receiver	100
receiver.lastname	String	Mandatory	Last name of the Receiver	100
receiver.middlename	String	Optional	Middle name of the receiver	100
receiver.mobileNumber	String	Conditional Mandatory for payments to Wallets	Receivers mobile number	12
receiver.email	String	Optional	Receivers email address	200
receiver.address.countryCode	String	Optional		3
receiver.address.postalcode	String	Optional		50
receiver.address.line1	String	Optional		300
receiver.address.line2	String	Optional		300
receiver.address.city	String	Optional		200

purposeofpayment	String	Mandatory	The purpose of payment or narrations	200
channel	String	Mandatory	http header	100
signature	String	Mandatory	Signature based from(rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber destination.BIC.wallet and channel	

Response: JSON
Response Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
reference_number	String	Conditional	Transaction reference number	50
quotationId	String	Mandatory	Transaction Id returned in Quotation response	
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment	10
source.currency	String	Mandatory	Source of funds currency	3
source.amount	String	Mandatory	Amount	10
source.accountNumber	String	Mandatory	Debit account	20
destination.currency	String	Mandatory	Source of funds currency	3
destination.amount	String	optional	Converted by the provider	10
destination.bicWallet	String	Mandatory	Destination Bank or Wallet	300
destination.accountNumber	String	Mandatory	Credit account or Mobile	50
charge.amount	String	Optional	The amount charged	
charge.currency	String	Optional	The currency of the charge fees	
status	String	Mandatory	POSTED PENDING REJECTED	20
responseCode	String	Mandatory	Any response or error code provided by Provider	20

responseDesc	String	Optional	If failure reason or addition date	300
transactionId	String	Conditional	If posted successful the Provide transaction id	50
channel	String		http header	100
signature	String	Mandatory	Signature based from (rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber , destination.BIC.wallet and channel	

3.0 Status Check API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/status

Request Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String		http header	100
signature	String	Mandatory	Signature based from(rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber and channel	

Response: JSON

Response Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
reference_number	String	Conditional	Transaction reference number	50
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment	10
source.currency	String	Mandatory	Source of funds currency	3
source.amount	String	Mandatory	Amount	10
source.accountNumber	String	Mandatory	Debit account	20
destination.currency	String	Mandatory	Source of funds currency	3
destination.amount	String	optional	Converted by the provider	10
destination.BICWallet	String	Mandatory	Destination Bank or Wallet	300
destination.accountNumber	String	Mandatory	Credit account or Mobile	50
fee.amount	String	Optional	The amount charged	
fee.currency	String	Optional	The currency of the charge fees	
status	String	Mandatory	POSTED PENDING REJECTED	20
responseCode	String	Mandatory	Any response or error code provided by Provider	20
responseDesc	String	Optional	If failure reason or addition date	300
transactionId	String	Conditional	If posted successful the Provide transaction id	50
channel	String		http header	100
signature	String	Mandatory	Signature based from (rrn, source.currency, source.amount, source.accountNumber, destination.currency, destination.accountNumber , destination.BIC.wallet and channel	

4.0 Country Names / Codes lookup API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/country-query

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String	Mandatory	HTTP header	100
signature	String		Signature based from(rrn, transaction_type and channel	
region	String	Optional	Static, possible values AFRICA,AMERICA,EUROPE,ASIA and ALL Default ALL	100

Response: JSON

Fields:

Response: JSON

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String	Mandatory	HTTP header	100
signature	String	Mandatory	Signature based from(rrn, transaction_type and channel	
region	String	Optional	Static, possible values AFRICA,AMERICA,EUROPE,ASIA and ALL Default ALL	100

Response code	String	Mandatory	00 Success 1. Failed System error 2. Not found	5
countries.country_code	String	Optional	Two characters Country code ISO 3166-2	10
countries.country_name	String	Optional	Full country names	300

5.0 Bank Name / Code lookup API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/bank-query

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String		HTTP header	100
signature	String	Mandatory	Signature based from(rrn, transaction_type and channel	
country_code	String	optional	Two characters Country code ISO 3166-2	10

Response: JSON

Fields:

Response: JSON

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12

mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String		HTTP header	100
signature	String	Mandatory	Signature based from(rrn, transaction_type and channel	
country_code	String	optional	Two characters Country code ISO 3166-2	10
Response code	String	mandatory	00 Success 1. Failed System error 2. Not found	5
banks.bank_code	String	Optional	Bank BIC	100
banks.bank_name	String	Optional	Full bank Name	300

6.0 Wallet Name / Code lookup API

Method: POST

Request: JSON

Endpoint: /api/outward/v1/c2c/wallet-query

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String		HTTP header	100
signature	String	Mandatory	Signature based from(rrn, transaction_type and channel	

country_code	String	optional	Two characters Country code ISO 3166-2	10
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Response: JSON

Fields:

Response: JSON

Fields:

Field/Tag	Type	Requirement	Remarks	Max Length
rrn	String	Mandatory	Unique transaction tracing number	12
mode	String	Optional	Preferred Provider, Can be defaulted or auto selected	50
transaction_type	String	Mandatory	Mode of Payment (C2C C2B B2C)	10
channel	String		http header	100
signature	String	Mandatory	Signature based from(rrn, transaction_type and channel	
country_code	String	optional	Two characters Country code ISO 3166-2	10
Response code	String	mandatory	00 Success 1. Failed System error 2. Not found	5
wallets.wallet_code	String	Optional	Wallet short name	100
wallets.wallet_name	String	Optional	Wallets full name	300

7.0 Java Signature Generation

Steps to generate signature.

1. Concatenate the fields described as part of the signature.
2. Sign the resulting text with a Private Key
3. Encode resulting binary data with Base 64.
4. Put the encoded data in signature field

Below is a sample Java code:

```
public static byte[] signData(byte[] data) throws Exception {
    Security.addProvider(new BouncyCastleProvider());
    KeyStore ks = KeyStore.getInstance(KEYSTORE_INSTANCE);
    ks.load(new FileInputStream(KEYSTORE_FILE),
KEYSTORE_PWD.toCharArray());
    Key key = ks.getKey(KEYSTORE_ALIAS, KEYSTORE_PWD.toCharArray());
    PrivateKey privKey = (PrivateKey) key;
    Signature signature = Signature.getInstance("SHA256withRSA", "BC");
    signature.initSign(privKey);
    signature.update(data);
    return (signature.sign());
}
```

In Java signature would be calculated as follows:

```
//Signature fields (rrn, source.currency, source.amount,
source.accountNumber, destination.currency,
destination.accountNumber ,destination.BIC.wallet and channel)
String clear = request.getRrn()+request.getSource().getCurrency()
    +request.getSource().getAmount()
+request.getSource().getAccountNumber()
    +request.getDestination().getCurrency()
+request.getDestination().getAccountNumber()
    +request.getDestination().getBICWallet()+request.getChannel();
byte[] signedData = signData(clear.getBytes());
Base64 base64 = new Base64();
String signedEncoded = base64.encodeAsString(signedData);
```

8.0 JSON Samples

Quotation sample request

```
{
  "rrn": "4543545",
  "reference_number": "454545",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "source": {
    "currency": "KES",
    "amount": "1000",
```

```

    "accountNumber": "132232324"
  },
  "destination": {
    "countryCode": "GH",
    "bicWallet": "BIXXXX",
    "currency": "GHS",
    "amount": "",
    "accountNumber": "343453454354"
  },
  "channel": "STK",
  "signature": "fsdfdsfdsfsdfsfd"
}

```

Quotation sample response

```

{
  "rrn": "4543545",
  "reference_number": "454545",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "source": {
    "currency": "KES",
    "amount": "1000",
    "accountNumber": "132232324"
  },
  "destination": {
    "currency": "GHS",
    "amount": "3000",
    "accountNumber": "343453454354",
    "bicWallet": "BIXXXX"
  },
  "status": "CREATED",
  "responseCode": "00",
  "responseDesc": "Quotation Created",
  "transactionId": "BMX4656GH",

```

```
"channel": "STK",
"charge": {
  "amount": "20",
  "currency": "GHS"
},
"signature": "w432432432432432432432423"
}
```

Sample Transaction Confirm request

```
{
  "rrn": "4543545",
  "reference_number": "454545",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "source": {
    "currency": "KES",
    "amount": "1000",
    "accountNumber": "132232324"
  },
  "sender": {
    "firstName": "John",
    "middleName": "",
    "lastName": "Donald",
    "mobileNumber": "",
    "email": "",
    "address": {
      "countryCode": "KE",
      "postalcode": "",
      "line1": "",
      "line2": "",
      "city": ""
    }
  },
  "destination": {
```

```

    "currency": "GHS",
    "amount": "",
    "accountNumber": "dsfdfdf",
    "BICWallet": "BIXXXXX"
  },
  "receiver": {
    "firstname": "DOE",
    "lastname": "JOHN",
    "middlename": "",
    "mobileNumber": "",
    "email": "",
    "address": {
      "countryCode": "",
      "postalcode": "",
      "line1": "",
      "line2": "",
      "city": ""
    }
  },
  "purposeofpayment": "FEE PAYMENT",
  "channel": "STK",
  "signature": "xcvbbggngngnnn"
}

```

Transaction Confirm response

```

{
  "rrn": "",
  "reference_number": "",
  "mode": "",
  "transaction_type": "",
  "source": {
    "currency": "",
    "amount": "",
    "accountNumber": ""
  },
  "destination": {
    "currency": "",

```

```

    "amount": "",
    "bicWallet": "",
    "accountNumber": ""
  },
  "charge": {
    "amount": "",
    "currency": ""
  },
  "status": "",
  "responseCode": "",
  "responseDesc": "",
  "transactionId": "",
  "channel": "",
  "signature": ""
}

```

Country Lookup Samples

Request:

```

{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "region": "ALL"
}

```

Response:

```

{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "region": "ALL",
  "countries": [
    {"country_code": "US", "country_name": "United States of America"},
    {"country_code": "UK", "country_name": "United Kingdom"},
    {"country_code": "CN", "country_name": "Peoples republic of China"}
  ]
}

```

Banks lookup samples

Request:

```
{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "country_code": "US"
}
```

Response:

```
{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "country_code": "US",
  "banks": [
    { "bank_code": "CITIUSNEW45", "bank_name": "Citibank US New york"},
    { "bank_code": "BARUKBO99", "bank_name": "Barclays US Boston"}
  ]
}
```

Wallet lookup sample**Request:**

```
{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "country_code": "TZ"
}
```

Response:

```
{
  "rrn": "1481184321405",
  "mode": "THUNES",
  "transaction_type": "C2C",
  "signature": "UlgQakRKG0yLVG673P5hg1os2DgGJMB2v1SY7TpeUXh3pfJDp5fLcLw==",
  "country_code": "TZ",
  "wallets": [
    { "wallet_code": "MPESATZ", "wallet_name": "Vodacom Mpesa Tanzania"},
    { "wallet_code": "TIGOTZ", "wallet_name": "Tigo Pesa Tanzania"}
  ]
}
```

8.1 XML samples**Quotation request**

```
<?xml version="1.0" encoding="UTF-8" ?>
<QuotationReq>
  <rrn>1481184321405</rrn>
  <reference_number>HNKM900353454354J</reference_number>
  <mode>THUNES</mode>
```

```
<transaction_type>C2C</transaction_type>
<source>
  <currency>KES</currency>
  <amount>1000</amount>
  <accountNumber>34353454545</accountNumber>
</source>
<destination>
  <countryCode>US</countryCode>
  <BICWallet>CTNY890</BICWallet>
  <currency>USD</currency>
  <amount>90</amount>
  <accountNumber>4225875254</accountNumber>
</destination>
<channel>MFS</channel>
<signature></signature>
</QuotationReq>
```

Quotation Response

```
<?xml version="1.0" encoding="UTF-8" ?>
<QuotationResp>
  <rrn>1481184321405</rrn>
  <reference_number>HNKM900353454354J</reference_number>
  <mode>THUNES</mode>
  <transaction_type>C2C</transaction_type>
  <source>
    <currency>KES</currency>
    <amount>1000</amount>
    <accountNumber>34353454545</accountNumber>
  </source>
  <destination>
    <BICWallet>CTNY890</BICWallet>
    <currency>USD</currency>
    <amount>90</amount>
    <accountNumber>4225875254</accountNumber>
  </destination>
  <status>POSTED</status>
  <responseCode>00</responseCode>
  <responseDesc>POSTED Succesful</responseDesc>
  <transactionId>3433DF444CDE</transactionId>
  <channel>MFS</channel>
  <charge>
    <amount>10</amount>
    <currency>USD</currency>
  </charge>
  <signature></signature>
</QuotationResp>
```